

ActInf Livestream #033.0 ~ "Thinking like a State - Embodied intelligence in the deep history...."

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me [Music] hello everyone welcome to actinlab live stream number 33.0 it's november 18th 2021 welcome to the actin lab everyone we are a participatory online lab that is communicating and mute if you're not going to be speaking thank you we're a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this slide this is a recorded and an archived live stream so please provide us all with feedback so that we can improve our work all backgrounds and perspectives are welcome here we'll be following good video etiquette for live streams and respecting our awesome facilitator blue at this short link you can find the streams past and upcoming and we are about to be heading into the second half of november 2021 where on the 23rd and the 30th we'll have two participatory group discussions on this very paper we're going to discuss today thinking like a state and then we have yet to set the december papers so if you know someone who might be interested in joining or you yourself would like to discuss a paper of yours get in touch with us today in active stream 33.0 the goal is to learn and discuss this paper thinking embodied intelligence in the deep history of our collective minds by aval guinin kalu and the video like all zero videos even though we do have the author here it's just an introduction to some of the ideas and how to pronounce them not a review or a final word so we'll introduce ourselves and then go into aims and claims abstract and roadmap keywords figures and formalisms etc so let us know especially if you'd like to participate in the upcoming 33.1 and 33.2 we'll start with just an introduction of saying hello and what we're excited to talk about or resolve either in the dot zero or in the upcoming group discussions and then we'll culminate with the author and transfer facilitation over so i'm daniel i'm a researcher in california and i'm just excited to check in on the progress of this important topic people have been theorizing about societies and cities and states and everything in between since as long as they were there probably a little bit before so it's just a fun topic and i'm really excited to see how active inference can give a new lens at resolving some of the long-standing

questions in that area i'll pass it to stephen yeah thank you daniel um i'm steve and i'm based in toronto and uh i'm really interested in the multi-scale implications of state and niche construction particularly for me because i'm interested in working around what i call social topographies which is probably at the scale of events and moments in people's like lived experience and in their architectures and places so i i'm really excited that this is bringing in this idea of state and larger scale but also bridging that with active inference from the kind of faster smaller scale so yeah lots of good stuff here lots of ontological questions as well which i think is really cool um and uh i passed it over should i pass it over to blue blue or sharp yep over to you so i'm blue i'm an independent research consultant in new mexico and i really like this paper and i really like working and talking with avell about ideas because these ideas are kind of at a bigger scale i am interested in self-assembly of things like at the cellular and maybe even subcellular level and like this takes it to the very big level and i think that active inference really kind of unifies this it's a skill-free framework and so that's kind of what i'm excited to talk about today and maybe not resolve but discuss and get some other perspectives on um and i'll pass it to the first author of the paper or the only author of the paper avail if you'd like to just introduce yourself say a few words so i'm available i have a training in collective science and physics mainly i'm interested in philosophy and a lot of things what we learned here is that i funded and helped organize cairo's research participative citizens laboratory which aim is to understand how to act or to navigate contemporaries and show historical dynamics and part of this is to understand why we have states and how those states that we now have are like um so the goal of the paper is to basically understand how states have constituted and what it tells about their continued structural identity through ages and this of course the topic is big there are several kind of states but the very first thing we need to do is to understand what your state is and where it comes from and that's what i'm trying to do there awesome so moving on to the aims and claims of the paper uh so this paper thinking like a state history of our collective mind in the paper it states that the core argument of this article is that states can be and should be understood as hierarchical control systems essentially similar in their core physical architecture to brains and we will leverage active inference to ground the study of intelligence in the way an organism maintains their structural autonomy to formally assess the collective brain hypothesis and the big question of the article is what kind of intelligence is displayed by city-state systems and how is it embodied in their structure so we'll go through the abstract daniel do you want to take the first section of the abstract yeah stephen first oh just one comment just because i know you mentioned kairos there and i've

done an activity with the kairos blanket or it's like a mapping thing where you stand on parts of canada and to look where it came from so i don't that connects to i if i'm writing you kairos is working at that kind of regional national scale to look at questions around that and from what i hear that's informing how your research is being approached is that correct i did not understand the question um not correct yeah it was an activity it's an activity actually not to do with a fristen blanket literally where they put a big blanket um about parts of canada on the floor in a room and then people have to stand on it and they keep taking away bits which were basically um which which were not are no longer in the territories of the first nations and then you see how much is left there's not a lot left so to speak so it's talking to that idea of statehood and settlers and how much is left and i wonder if that speaks to the wider work of kairos and questions of states and how do we understand how we fit into that or maybe i'm i'm mis misunderstanding so that fits in the work of keros because it uh tries to bridge scales i never heard of these keros broadcast this is not related to what we do in any environment it helps embody an impression of how well settlers have appropriated indigenous american territories so this is something that's called that fits the mandate of karus but the core idea of keroses it's the greek notion uh for time for three in a greek and carlos is the instant uh and it's uh coded actively like it's an instant where you can see the opportunity and um this is basically what embodied agents do they build skills and they build capabilities and they enact them when it separates this is the goal of chaos to understand and reproduce uh this uh core ability and a lot of people have used kerosene you have anarchists that claim um that are called cairos you have a christian concept that is called cairos you have a lot of people that make the use of the concept because it's very broad and it's very um it's not something that has been imported in the christian concept of times really in like what you have today so if you want to talk about this you use you just use the greek word which is kairos so yep just one thought then i'll go to the abstract which is that kairos as opposed to kronos like a chronometer with decimal time kairos is like timeliness and it brings action and sort of a little bit of a moment that transcends the tick tick tick tick tick so it's a cool term i'll read the first third of the abstract this article aims to show how the deep history of early state societies entails the development of a collective form of cognitive agency it relates classical works in the anthropology of states in particular scots seeing like a state with the inactive account of biological and cognitive organization thanks to the unified ontology for self-organization dynamics across scales offered by the active inference framework stephen do you want to take the next piece of the abstract yes okay so active inference conceives of cognition a synchronization across

individuated sensory motor states it entails that biological or sociocultural constraints display minimal form of cognition by shaping the behavior of faster dynamics in a certain way when such constraints collectively define a basic life form an integrated operationally closed system they can therefore be said to embody adaptive knowledge properly speaking and then i'll just take the last part it says the inactive inference account i articulate here strongly motivates and methodologically grounds a hoist a holist approach in the social sciences indeed it grounds the study of human societies in the role of structural constraints whose meaning depends both on the broader systems activity and in the historical context of their emergence the present account of the dynamics of early urbanization and state genesis aims to illustrate this approach and a quick just walk through the road map of the paper we're gonna go through the introduction section and then there are three main sections of the article the embodied intelligence in nested minds is the first the second is the deep history of cities and states the third from the urban metabolism to the collective mind and then we'll go through the conclusion and first up in this dot zero contextualization video we're going to discuss the keywords which i think there were no official keywords but these are the words that i created for the paper when discussing it at the conference so we pulled out intelligence autonomy embody intelligence active inference the free energy principle the concept of thinking through other minds and we've done that in a stream before daniel is that true i think we have right um and then uh niche construction so i don't know i don't have the facts on the on the exact number of the thinking through other minds um stream but we can find that out somewhere during this during this talk we'll look it up evel do you want to maybe speak to we'll give you the first stab at talking about what you mean by intelligence and autonomy we've pulled a couple quotes out of the paper but maybe you'd like to so if i remember well intelligent etymologically it's the capacity to make links to connect dots of data and to build antiquity understanding of it i don't want to commit to any specific notion of intelligence because i am not aware of any meaningful first principle definition of it there is a lot of work in measuring intelligence like iq which maps to specific cognitive abilities and has literally zero reason to generalize to other life forms than humans and we contribute to other life forms than humans in occidental societies the national autonomy is much more specific it's important to uh okay so etymological is one's ability to build one's own rules so for example if your city state has laws and the organs that make the law are within these states it is autonomous in a strictest sense it is also recruited in the study of life and condition and the way cognition is grounded in life by inactivist scholars where it is related to a personal closure under precarious consensus

so it is something that is able to operate and able to recreate the conditions in which in which it operates while there is no uh exogenous force that help himself do so for example if you have a candle it helps there is a burning fire which uh generates a flow from season so as to maintain itself but if someone does not keep um fuel going this dies so this has no really way to recreate the conditions at which it operates this is in contrast to the simplest bacteria there is which is able to look for food basically so the active concept seeks to capture that nice um daniel do you maybe want to read the the quotes from the paper directly referring to intelligence and autonomy as described er intelligence is the general ability for a system to understand the world either in a reflexive propositional way or as enacted in their ecological activity and autonomy refers to the property of operational closure under precarious circumstance ie to the capability of biological systems to continuously maintain and recreate their own structural identity as a result of the collective activity of their constitutive processes right and these are processes that are actively generating and sustaining identity and i think avell spoke really well to operational closure um in the what you just described but we just want to refer anybody who's listening to this paper by montevelle and massio in 2015 this biological organization as a closure of constraints as closure of constraints so i want to emphasize that there is i think it's pretty old the percentage it has at least 10 years and the montevelle paper is an attempt to formalize it in uh logics so it is not necessarily the the reference on this i don't remember what is the reference paper on this awesome thank you um maybe let's move on to embodied intelligence and stephen if you uh this is perfect for you if you want to read the quote from the paper describing um what embodied intelligence refers to yes okay embodied intelligence refers to the ability for understanding adaptation so sort of a mixture of both that is imprinted in an agent's physical structure rather than in some general computational ability yeah that's significant and then we have the wood lice alias yeah don't use the jitsi chat fyi you can use key base if you want a back channel but that's the wood lice example that carl firsten provided in the 2018 example that's a little shout out there awesome yeah so the uh does anybody want to say a few more words about embodied intelligence maybe avel or steven or if there's something that comes up for you guys feel free to raise your hand if there's something else you want to discuss and the slides are the slides that we prepared avail for the talk they're in the key based chat as well so i'll i'll mention just to say i mean because we're kind of introducing at this stage i won't start raising those and loads of questions and we'll save that for the the next one but uh and i just note that um that this is useful just to highlight how this is being structured because you've got this word intelligence

embodiment and uh i know in the there's them talk about the brain or the idea of the brain and the idea of the mind um so i think it's interesting here that this this paper sort of bridges a bit across um the brain intelligence mind culture kind of uh area so when it's talking here it's kind of seems quite an open description and is um really making sure it's clear to speak to the difference to this general computational ability and that it's it's somehow a different type of uh knowing that's been done maybe with the word you know morphological computing sometimes is used in that in that world nice thank you um maybe daniel do you want to explain the free energy principle and active inference sure so as on the slide here the free energy principle states that dynamical systems given sufficiently regular boundary conditions which are required for their continued existence spontaneously self-organize so as to minimize their expected variational free energy through time effectively maximizing evidence for the world model they enact so the expected variational free energy which we talk about in more detail elsewhere it's kind of like all aspects of the system through time are what have to be handled in order for persistence using some technical formalisms that turns out to have equivalence with evidence for a generative model where one is persisting active inference explicitly brings action into that inference process and it's the process by which dynamical systems maximize their evidence for the model of the world which they enact this can occur by a system altering its environment to fit expectations i.e niche modification so it's the difference between the free energy principle merely saying that persistent systems are going to be isomorphic with those that have high evidence for the world model which they enact and then active inference as a corollary process theory says and the way that they're going to do that is through inference updating their internal model or through action which can be just sort of like a flock of birds action in the sky or it can be niche modification like stigma g um and then we can go on to thinking oh stephen you have a comment to say yeah just one piece here also sort of flags up for me is the the use of the word model because we talk about their own model of the world and yet um often we think of models as something that we've created in the world you know on a piece of paper and things like that so again there's a sort of ontological challenge in the sense that it's it is kind of like a self model but it's it's it's a self model that's not a representation um so um but our language is kind of couched in our history um so anyway i think that might be something that sort of resurfaces and have to kind of keep restating because people might have this representational kind of inbuilt way of you know those metaphors are quite embedded so it's hard for people to necessarily switch to what it means to move away from representation representational approaches i have one question as

well the reason why we're talking about active inference and free energy principle in this paper so avo what does active do for this paper are there other contender theories in the game how else are people trying to take a unified approach to studying the specifics and the generalities of states so the um the approach i have i define as an active inference with the n between parenthesis it's a synthesis between what is said by an activist and what is said back to influence theorists the core framework is inactive i'm interested in um characterizing states uh state societies as a network of constraints so as uh slow processes that are exist at the holy scale uh not with the individual not between individual but in some sense over individual and that shapes the activity of individuals so as to maintain itself that's the core argument society is our organism but now if i want to get there i need uh some strong instincts i if i need to get there in if i need to formalize the role of states which are at core information processing machines in it i need to formalize in a struggle in a stronger sense the relation between public synchronization basic authorization and epistemic states and that is precisely what active influence so i need uh the two uh i need elective uh formalism the native approach and i need active influence formalism to make the link between the ecological activity of states as an international process and the body the constraints that actually consists that actually statistics consist in we can i guess move on to uh thinking through other minds um stephen did you have a comment just one other comment as i was interested by the um you know you're using the term constraints and in active inference is sometimes thought of as the higher level processes are in some ways constraining the behavior of lower level processes rather than necessarily dictating them so they're you know it's look like i'm doing the faster jig and someone's just slowing me down a little bit um and so on so yeah just curious that that word constraints has brought in because often people think of you know intelligence is all about doing things and you know not constraining things so anyway just curious about that yep that's related to synergetics hawken not bucky and about like a wave and a ripple like the wave constrains where the ripple can go the faster fluctuations are constrained by the slower ordering parameters so here we have thinking through other minds and this is another body of work which we won't spend a ton of time on but for organisms that are making generative models of their niche sometimes that niche contains other organisms such as their self and it matters how those other organisms also include generative models of oneself so thinking through other minds is describing how in a shared epistemic niche shared informational niche with shared salience cues for example that two different agents are able to coordinate because they are thinking through each other's minds like if i put up my hand then blue will think that i

want to speak so it's sort of like a little bit of a theory of mind but with more of a act imp flavor so both like the idea of constraint and this concept of thinking through other minds bring me back to mike levin's paper the computational boundary of the self because of of the nested like cognitive cones right and when he kind of talked about it he talked about you know the the higher order or like the larger level the higher level organism um or collective constrains the behavior of like the the individual um like sub level collectives um so that was kind of this idea of constraint and and then also um like thinking through other minds like specifically we talked about how connections between cells individuals but the connections that that essentially breed information sharing so so when you have like a shared informational niche that enables this this process kind of of thinking through other minds where you think more as a collective and so that expands the cognitive boundary of the self to then include other you know individuals within that cognitive boundary stephen yeah and that ties into the work of andy clark and uh the extended mind so there's a um so the way that that gets uh there's there's a bit of um talk about how do you think about that extension happening and how much is it an extension of the individual and how much i suppose is what's being talked about here is how much does what has been extended start to become its own sort of cognitive agent nice and maybe very appropriately avail has returned and we are moving on to talk about niche construction and i think um avel might have a lot to say about the niche and its role in extending the cognitive boundary um so maybe do you want to give us a few a few insights into can we help us think through your mind modify the audio visual i'm not sure what i could say about the niche sorry modify the audio visual niche of our joint well that's attention we modified the niche i will actually actually within english i don't know anyway um i can't change your model of the niche this is not what i'm going to do i think i'm not trying to do that activity at least uh whatever um so the there is body of work in biology uh evolutionary biology which is centered on how organisms modify their direct environment in a way that somehow fits their their ecological needs so a typical example is how beavers will be dumb source to get i i honestly don't know what beavers get out of them but they get something the basic arguments uh in the um thinking through of the minds uh framework is that the way humanifier niche has a cognitive value it's externalized expectations by analyzing expectations it allows expectations to synchronize between individuals there is a very clear analogy or identity between once you accept there is a cognitive value in a niche construction as the externalization of expectations there is a very clear analogy and even identity between uh niche construction and excellent cognitions beaver are just extending uh their cognition their

expectations the domain of the world that somehow participates to the fact that they um they live they keep on leaving and understanding the world there is a pretty basic way in which an organism can expand the extent of the world the domain which is operationally closed and which participates actively in their maintenance this is basically what's there is to be found in this construction by shaping their niche organism shapes their world they shape their cognition they shape their body i'm not sure i'm trying to give a general insight in uh initial section and how it relates but i'm not sure relevant it is because i'm not sure what is the point that is to be made followingly just that niche construction is an important mechanism of extended cognition like the recent stigmergy models show that individuals don't need to carry on board or internally the entire cognitive apparatus they only need to know how to interact and potentially modify a niche and then also i think this is important when we're talking about cities and so on and states because our niche can as per thinking through other minds include other others like us or distant from us but also it includes the abiotic niche so climate change and resource utilization those are niche modifications there are some radioactive mountains that have been niche modified and so it it provides a continuity between the kinds of ecological changes that humans make with the cognitive approach to ecosystems and so that's why it's a very rich framework to build off so i just want to read this quote from the paper really quick and then and then stephen will come to your comment so it says as stated previously the network of constraints defining the structural identity of basic life needs not be printed in flesh it only needs to influence faster processes so as to continually recreate itself and there the references for this are there on the slides if you care to probe further it says such constraints can be embedded in the informationally rich sociocultural niche which enable the human ecology by affording adaptive coordination and the development of complex skills and i i think one of the the main things that um this paper does and uh feel free to correct me if i'm wrong but it kind of takes the idea of cumulative cultural evolution brings the niche into this as a cognitive extension so i think that's one of the key moves made here basically i know a cultural revolution uh i have known revolution for like five years or so actually in france for one year only so cultural relationships so i see things and i'm currently writing a paper on uh that could be integrated on reductionism in cultural and the fact is cumulative cultural evolution in cultural revolution is the replacement of low fitness variants by high fitness variants within a variant i mean of cultural tokens that are implemented within minds our behavior because there is not a strong ontology on what the replicator is in cultural evolution a basic inside world to convey is that culture does not stop in the brain culture does not certainly

does not stop in uh individuated cultural cocaines one of the goal of the paper is to show that social revolution it has analyst extension uh our intention i don't know it does it exists uh not only within the minds of individual and within uh the verbal effects of cultural tokens it exists instructors that are not apprehensible uh by an individual and that are not entailed by any specific cultural token hence the holism thanks thank you stephen yeah also bringing that beaver example is to some extent the beaver constructs a dam and to some extent maybe it grows a dam um maybe in the same you know so by adding pieces it's it's growing something a bit um and at what point does it sit stand back and say oh that move that one a bit to the left move that a little bit to the right and sort of improve the construction and how much is it an evolving process i suppose there's also the idea that once something has grown and it's changed the niche um something else you know just like the brain effectively has to grow before it can do its thinking um so maybe there's something about how these structures have grown and been constructed so i don't know i won't i'm just mentioning that awesome thank you um i guess uh maybe we kind of got ahead of our of ourselves uh talking about niche modification and cultural evolution um i love this picture because it really uh shows like kind of um how much we have offloaded our our cognitive selves into the niche environment like we don't need to know how to use directions or what restaurants are good or we don't need to know actually any of that anymore because it's all like riding around with us in a smart phone in our pocket all the time um so so we've just put put our brain we've offloaded all of the cognitive capacity of our of our brain or at least a good portion of it so what do we think about now what makes me think about that always um it says in the paper we claim that niche modification is critical for cumulative cultural evolution and advancements such as writing allow active inference to occur between agents and stigmatic cues embedded in the niche although did i say what you did i write what you just said what i want you to note is that what you described the way we offload the information on our social analytical and technological environments is the classical uh collective brain picture which is entailed by uh mostly um void which has shown in handwritten of cumulative cultural evolution uh what i want to show is that uh together step beyond we do not only have a self that is augmented by environmental cues there is a self outside ourselves there is uh things that shape social life and that are not their agency that shape social life and that is not human that is institutional so that's the step beyond the collective brain in the vanilla a collective brain view which is maybe an anti-collective brain there is actually such a thing as a connective brain starts an image and we can study it well we can study it now through wikipedia and the internet like our network of the

collective brain steven yeah it's interesting as well in terms we have all this extended information but there's this there's a very good book called the paradox of choice so the the thing is we have all this information but you still have to do an action policy on what to do and the challenge is you can have so many you know it's like going to the supermarket and you know you can buy 50 different types of olive oil okay it's just olive oil at the end of the day but now you've got to choose between 50 and you've got you know so this there's a sense of abundance but there's also a need for an action policy which is of course what active influence talks about um that ultimately it's not the volume of information that's going to be the game it's the ability to act and uh inappropriate ways um and make sense of how to be in your niche so that's kind of interesting um so we are gonna move on um we're gonna go into the next section embodied intelligence in nested minds but but just to quickly summarize here like what was the main the brought up in the introduction so we've gone through the keywords but just to clearly just state it out autonomous systems are defined by a set of processes that create and sustain their identity they embody their understanding and adaptive abilities within their physical structure they enact a model of their world and seek to maximize evidence for this model through active inference and they can maximize model evidence through modification of their niche so just to kind of sum up what an autonomous system is and what where we are at this point and then we'll move on to um discussing embodied intelligence in nested minds so a quote from the paper it says uh here we argue that innovations are instead an emergent property of our species cultural learning abilities applied within our societies and social networks our societies and social networks act as collective brains we outline how many human brains how many human brains which evolved primarily for the acquisition of culture together beget a collective brain oh this isn't actually from from the thinking like a state paper this is from the innovation in the collective brain paper and this framework this collective brain framework grounds the study of intelligence in the way that an organism maintains structural autonomy and allows the formal assessment of the collective brain hypothesis so the collective brain hypothesis has been put forward but not formally assessed up until this point avail do you want to talk a little bit more about that because i probably just botched it about how the collective brain pathogens is not commonly assessed yeah in cultural evolution you have what i'd say is the dominant theory is gen coevolution uh gene culture qualification theory which is basically centered on how um we have cultural variants that have an adaptive role and there are a set of cognitive bias or cognitive mechanism but actually cognitive bias uh bias actually actually that help us identify what are the most adaptive cultural variants and this

uh has a feedback effect on gene selection which has an effect on cultural selection and basically all of your interpretation is until the process of a convolutional process between a genetic and cultural evolutionary information processes the fact is there are claims there are numerous claims in the literature that there is some sort of collective computation that you entered our corrective cognition that is entailed by this so there is if i remember well uh henry and mata krishna who wrote the um innovation in the collective brain paper and there is i think boyd and um svaldino boyd which is basically one of the two founders of the initial formalism and smart dino which one of the most competitively study a member of the current literature that wrote a paper on a collective cognition no collective comp distributed computation because revolution has discrete computation i don't know neglect data but it's uh i want to accentuate it a lot it's words like they have a model of individual bias in a selection and they have a narrative interpretation of this somehow in telling some kind of computation and there is nothing in between there is a few paper less introductors that's kind of interested in the convolution of social networks and cultural variants access and smaller to name them there is literally not one paper that explains how is this supposed to be cognition what are the mechanisms what is the cognitive architecture there is nothing on it uh or it's marginal and i did not heard of it the goal of the paper uh is to basically explain how we can grant a view of connective mechanisms that are entitled by the social cultural networks and structural constraints that are embedded within a society and that's it's not a model because uh to build a full causal model of any brain or organism is not tractable but it's to prove that it's uh meaningful it's a there is a positive way in which we can model societies as or at least process within societies as a collective form of and a key example of this is the state because we have access to the mechanisms we have access to different papers with which they understand and predict and control reality awesome that's a perfect i think transition into the next slide do you want to read this first daniel this paper uses the framework described to illustrate that the co-evolution of top-down control and cultural niche construction enabled the inversions of the city-state complex and how these collective organizations maintain structural autonomy and so that's kind of the the summary i say of the of the section the embodied intelligence and nested minds uh section does anybody want to give a thought or we can go on to the next section just what you just said about how we know the mechanism level of the state i'm thinking we see its neurotransmitters and its action selection policies so it's like every system has advantages and disadvantages for its study this is one where many of the key features are designed and so it's a lot more like investigating the connections of a program that was written rather than the

potentially like map territory debates over like what is the bacteria really doing are we really measuring the bacteria but it's like we know how certain processes in the state happen but not all of them and citizens usually don't know steven yeah so following on to what daniel said a quick question in a way is much you may be trying to have a sort of a realist approach on the idea of the brain or being a brain or how much is it an instrumental tool or a metaphor so be curious as you know you take that correlation because you've got brain you've got mind you've got intelligence you've got extended cognition is it that you are really wanting to make it say that is like a brain that's kind of in enacting its own action policies or um so that would be that might be helpful so i think actually like maybe we should hold that to the end section from the from the because i think we're going to get more into how is a city like a brain um as we get closer to the end of this contextualization we'll save it for dot one and dot two because we'll cover just the background here but it's a very good question yeah very good um it's going to provoke a lot of discussion so that's why i'm like well maybe we should get through the rest of the rest of the talk going on first so um it is so we're moving on now to the deep history of cities and states uh here is another quote from the paper uh stephen do you want to read this for us cultural niche construction allowed humans to alter their social cognitive ecology in a way that allowed extensive functional specialization and the evolution of supra individual agency through mediating institutions that makes me think about the evolution of functional specialization in ant colonies and it's like you can't have the super specialized task you know you can't have somebody who specializes on just sourdough bread until there's somebody else who can help them with some other tasks and that's the economic inspiration for why they call that process in ant's division of labor downstream of adam smith 1776 the economic division of labor and the candle maker and the shoemaker and all of that so once you have a lot of people living together they can specialize in a new way there are new institutions needed needed to mediate and that results in more being different i love how daniel like says ants and gets like giddy and excited a stephen yeah and i noticed this term super individual so there's kind of this um i suppose in the anchor you think of the queen but in in our world we we have these and actually i've been thinking about a lot of this myself in terms of how we relate to these bigger scales so this idea of a supra individual or supra collective ties into the idea of the state you know because you know we think of you know the word state is actually used literally in the united states to be you know a region but there is an idea that is trying to maintain something so um that term i think is a is a useful one to sort of put out and uh and that is super individual agency so that ties it into active inference in a in an interesting way um

it would be interesting to know the term socio-cognitive ecology it's not one i've heard before in quite that way so i don't know that could be something just to just to maybe ground that term in this context well we can return to it and i added the book cover for leviathan with the person made up of other people in this section of the paper avelle explains how state niche niche construction depended on serial monoculture uh the technology of writing material infrastructure uh dense sentiment dense settlements and engineering societies from the top down as documented in seeing like a state by scott um avall do you have a few maybe words to say about this i mean more than the brief kind of overview i just gave so i'm not sure identified a key uh subject like on the um importance on the uh morphogenesis of uh states from a social scientific perspective just we're just talking about state niche construction like what what did that depend on all of the things but the career states is um i think we could have a lot of definitions uh their campaigns it i think it's okay it is said in certain texts that it is uh essentially a monopoly of on legitimate violence so if there is a group of men that has weapons and that has the exclusive rights to use it and that this right is probably accepted it is a state uh i'm mostly interested in the mode of operation of a state so for me it's a chain of command basically it's an emissive systems that makes people do things so these those two definitions they entail the loss of human autonomy then tell that you have at least a subset of humans that are constrained to follow the laws laws that are uh extremely imposed so that's in clear deviation from what is done say in iroquois society where at least by the personal contact mostly no one has the power to coerce anyone else and any decision is grounded on rational debates so this is not what a state is like a state is when you are a guy with the biggest bird and you can say okay you go kill this person because i want to i want this weapon and you have a definition of the states that's uh sorry your definition of the state there is that is uh grounded in the sub-brain i'm not sure when it initiates i think it's with leviton which i did not read but basically that the state is a central actor that has power um on anyone which is um sorry of chats things so state is uh something that has power on anyone it's not tied to a specific so that entails the capability for a set of for institution or a set of individuals to basically coerce other people to do and this entail for example enough at least a north centralization of resources for other people to not flee or enough imbalance in wealth for other people to get to you even though they know it will strip them of their autonomy so it's about the grounding on which the literature that project literature on um states and state emergence is written and this entails agriculture uh trade a lot of ways for basically wealth to be centralized and powerful people to be able to constrain everyone else awesome and that like speaks very

well to um how states get a grip on their subjects and themselves um and what is the the common grounding maybe that um is used to get like to get a heterogeneous understanding of states oh sorry emergency alert ah your attention please your attention please update in action policy requested from the state exactly exactly so so now states get a grip on us through like our our cognitive offloading into the knee strike um stephen did you want to say something yeah i mean this also speaks to that challenge of being able to get a grip on things which are beyond the adjacent scale like that you know they're trying to get you know they have to use they have these technologies they can reach out to everyone but at the same time it's kind of they can't easily get a grip so there's um there's some interesting points there around um you know because there's a few scales between us and the state right and that's that's interesting and technological development little tiny chips that track where you move that makes it a little easier to get a grip on where somebody is moving so it's a co-evolution with the states and technology going back to agriculture and writing and ledgers and now we're in a new phase of ledgers from zero one two three nice stephen yeah and of course the challenge is that they get a grip on what you're doing but they their influence is about why you're doing it decontextualized so they think someone's doing x or y and it may be wildly different you know they may be underestimating the risk or they may be radically overestimating the risk of behavior awesome so i'm going to just flip through these next few slides uh dense settlements the what the function of the dense settlement did was allowed for economic codependency and we talked earlier about the functional specialization that was like you know necessary for this economic co-dependency and it also facilitated um the development of nested social communities uh um and daniel wrote is their scaling relationship here and i think we have to wait and address that maybe in the in the dot one and uh moving on it says enforcing state control over urban systems mandates building acceptance into the population and this is like how do you let the state be in charge of you how do you grant grip to the state and this is the people must perceive competency in the authoritarian regime and there's a redistribution of wealth like for example paying taxes and also symbolic legitimacy which is achieved through the church um largely or maybe maybe there's other ways but but was in in the in the early development of cities and states largely um religious based so what um there's a lot of things that that we've talked about so that tie into formation of cities and states and it's large-scale coordination collective crop redistribution which i thought was interesting like some people grow it some people eat it some people take it some people sell it and then the functional specialization specialized rules and the control of the money and the flow of the money so all

these things together unify to give a state grip and allow this kind of top-down control is is there any comments on that or if not we can move on to the next section just that if those are the necessary and sufficient features when there's a crisis of competency or distribution of wealth or symbolic legitimacy what is that forbode avail do you have a an answer i'm sure waiting and waiting in the wings yes uh the answer is basically uh a structural structural demographic theory which is central side results in clevelandics which basically states afford the possibility for the elites to ignore the social maintenance and just capture wealth when they do too much it causes political instabilities so and this entails uh fluctuating um protecting cycles of integration and distribution in states developments and by the way we in the sense of france the us and likely global capitalism are in a phase of disintegration which uh needs which entails that we have uh you don't need to understand how to organize outside or in spite of administration and to understand where they come from cool so we're going to move into the last section of the paper which is from the urban metabolism to the collective mind and we're going to go back to the main question of the article which was what kind of intelligence is displayed by city-state systems and how is it embodied within their structure and uh maybe stephen do you want to read this quote from the paper okay so answering this question in terms of inactive inference entails revisiting our accounts of the urbanization dynamics as the results of city-state systems actively managing to get a grip on their world ie energy cool and there are a couple more um quotes from this section uh that we can move on to daniel do you want to read this first one here basic autonomy is indeed formally defined as operational closure or more precisely the capability of a network of constraints to canalize faster processes like metabolic flows into maintaining and recreating itself i like that's kind of a twist on the kantian organism concept the teleological definition the organism is like the level it makes sense at and this takes a more operational and mechanistic approach towards closure of mechanism and constraints and autopoiesis rather than merely end directedness which doesn't immediately suggest a mechanism and then since the material and socio-cultural landscapes of urban systems canalize human activities so as to maintain itself the process of urbanization clearly intains the constitution of cities as basic life food comes in waste goes out city's alive awesome and i love this picture of like the elephant is the city um sir did you want to say i want you to something out that the first point you made is that um with the mountable and moisture definition or my account of them is a twist on the conscience idea that organism is the level at which it makes sense and it's basically the movement that monteville and monsieur and moreno make uh in activism at first it's surrounded in concertal philosophy and feminine

this this is the grounding in which uh varela and maybe machrena i'm not sure articulated uh the view that cognition is in fact grounded in life and what uh mostly people and the basque center for life sciences i think it's called try to do is to mechanize it to personalize it to ground it into specific uh chemical biochemical processes so it's you precisely captured what this account is meant to do and good good job inferring daniel based on based on the um you know stigmatic cues left in the niche uh and and we can move on to a couple more quotes from the paper um this is one that really resonated with me and it's paraphrasing jablonka and lam in 2006 this paper that's here the evolution of information in the major transitions says a full-fledged evolutionary transition typically entails the emergence of new forms of information systems mediating regulation at the collective level and or the heritability of collective traits and then another quote from the paper is also taken from this gaudian crawl paper in 2014 it says the emergence of city-states with their professional classes and their mediating institutions based on writing technologies constitute a clear case of such a transition especially in the case of the late agricultural civilizations and i i totally get this and also it really speaks to how stigmatic cues left in the niche environment can contribute to the formation of our collective brain i mean writing like we can leave stigmatic cues for generations yet to follow and some writing in stone right like so this is like still we can read it many many many many ages and generations later which i thought that was really uh poignant stephen yeah one thing i'd be interested in in terms of the state is um how much different areas of society or the state are trying to achieve similar types of outcome so when there's um and when there's i mean we've just had that case of um that video of a cortez being murdered and uh you know so like within the state you have internal conflicts you know within within the state itself um but also how much does the people at different scales have to be aligned or how much will the state itself kind of do what states do so to speak as its own mind awesome thanks daniel yeah i agree about not every single action in the state necessarily being the same like if you looked at the body you said well no we're supposed to be making glucose why are there cells that are breaking it down or like there can be sections that are doing processes that seem directly opposite to each other but that can be part of a broader regulatory process so brings into question you know what are the official processes versus the things that are in the gray zone of the state and how do we go from this model into more useful forms nice thanks um and so we can move now to another quote from the paper that is from paraphrasing this paper that we actually just did in a recent live stream um how to count biological minds by matthew sims and in in the thinking like a state paper it says if the nested hierarchy of regulation systems

enables adaptive behavior at the system scale it effectively enacts a collective mind regardless of whether system components retain their cognitive ability and this speaks back to what we were saying earlier about the the top level kind of constraining the lower levels like maybe there's like a uh once you contribute to the collective then you know you've contributed to something greater than yourself and your your own cognitive ability perhaps doesn't matter so much any thoughts on that or we can move on to some more topics from the paper daniel do you want to read this quote the closest analog of our intuitive picture of a mind in urban systems is their most disembodied mechanism for adaptive regulation administration indeed just like neuronal cognition it entails the top-down regulation by a specialized population with the help of a specific purpose information system it makes me think about those tubes that they used to put the physical packages in and send them and now they're special information channels but you better be listening when that channel calls so that's like futurama like they have like i mean the bureaucracy like the administration and i mean administration i love this quote from the paper i do think like administration is totally like disembodied from like reality and bureaucracy is like so imposing but like in the futurama they all have like you know it's like space age modern transmission like you could just click docuSign on the internet and like yet they're using like tubes literally like to send the things off um and it takes seven years to get approval or something ridiculous like that but it like it does work like that right bureaucracy is is kind of crazy this next quote says the institution of state-sponsored legal codes affords a way to directly transcribe administrative expectations into individual minds rule of law is therefore a core tool by which states create their world through building the basis of their own intelligent behavior into their socio-cultural niche and this is brilliant right like just yesterday my landlord called the codes officer um because my neighbors have weeds right like i know they're not allowed to have weeds it's beyond the expectation of of like you know our neighborhood that nobody has weeds and everyone knows that this is the expectation some people don't maintain their weeds and we all have the right and duty and obligation to you know impose that on our niche environment daniel um so the uh expectations the normative expectations a lot of laws are like you know it shall be within two weeks after this giving notice that this happens so their expectations and then they're called codes and so maybe think about predictive coding and it's like we can predict that this technology is going to be used this way so i know it's not exactly what people mean with the neural rate codes etc with predictive processing predictive coding laws are normative they're saying what should happen and that helps behave as expected you want to be on the right side of the road driving in the u.s so

that's like an expectation and then it becomes an intersubjective preference that is sponsored by the state so it's just kind of cool again showing points of contact between people's day-to-day life and their intuitive explanations for how these systems work and then it's like yes we have an ontology that picks up right with you goes a little further than you might have expected to awesome serval yes that's exactly the idea knows until the externalization of expectations the fact that they are normative is i think incidental accidental it's a phenomenon to their core role which is to shape behavior you have a project in legal studies i don't know how you call that which is called i think homogeneous ridiculous or homogeneous or some closely associated formulation which has a behavioral approach to bone we need to do that more yeah sounds good um so we just have some summary um quotes maybe from the paper and to kind of summarize the ideas that were presented here daniel do you want to read one or some of these okay i'm just picking which one i would want to read i think the second one is a lot like what we were just talking about the state defining expectations and then modifying its own niche that operational closure to implement those expectations lest it dissolve then i i like this top quote our notion of intelligence is based upon active inference a scale-free framework applying to any system with individuated boundary states and then it's like even the word state is upon as avail pointed out before it's at being state and french and we talk about internal external blanket states boundary states that sounds like it's going to be at the boundary of the empire and then this is what it looks like to apply active inference to a real system to go from a sort of bayesian graph boundary notion to being like well what if that boundary state country was intermediating and insulating between these other two so i like how even the word state comes into play i like that i i really like that too and i like that like that that's kind of a play on um like the first in particular physics like specific particular physics but also like physics as related to a particle um and and the idea of a state actually something when i was proofreading this paper was like you know i know that you're talking about like city states but it's not immediately obvious to the reader that like you might want to say like city states to kind of flesh that out like from the get-go because it is it can be such a play on word right it's a boundary state it's a blanket state it's an active state it's a you know city state daniel maybe one other thing we can talk about in the dot one and two is states as we know them are new and so it's only been a few generations like your great great great grandparents that's a long time ago it's a lot of tacos ago but it's a blink of the eye in terms of evolution and in terms of deep history and so even pre westphalia that was not so long ago and there's so many world traditions and so just realizing that we're in a very non-stationary moment with the

development and the implementation of states it's really important because it can be easy to be like oh it's a grind all the states in the u.n and you know 50 states in the u.s it's like even 70 years ago there weren't that many states in the u.s and there was different countries in the world so things are changing it just sometimes it's at a different time scale which makes sense because they're bigger and slower entities than we are cheval i want to point out two things first the pawn is not intended but no i will claim it is and no one will check um the second is that the received you is that states emerge with agriculture it is what gordy and kroll said in the paper i cited um it is not accurate or we don't know whether it separates uh you have a pretty good single review of the literature of the state emergence and the fact is we do not know what human societies were like before written history we do not know so i and most importantly it's um something uh grabber and one grow to a leading anthropologist in historical apology like to uh rave about uh there are kinds of conversations that are authoritarian there are kinds of transactions that are not so even with if we are signs of i don't know great ritual states or anything we don't know whether those kind of organization were like what we know today because uh western national states are extreme in kind of extractive and corrective behavior but they are shaped by war within europe and between europe and um colonial colonialized uh countries uh so i prefer i much prefer to ground uh the history in which i talk uh in europe and history in a modern history after westfalia after the birth or in the roots of the birth of the national states because it's pretty hard to talk about the roots of sumerian states we need to do so because those are states we need to do so if we want to talk about what states are but if you want to talk about what our states are we need to go uh in west faria we didn't mean not to go in summer or egypt or whatever so just to kind of play off of that um extractive and corrosive behavior and that states are often born out of war um something that you mention in the paper but um i i don't think maybe gets enough um attention is the concept of shared identity because in this shared identity like that's what makes us faction off into our own state we're going to divide and become something new and daniel was saying how we have so many states now so many more states than we've ever had before and this like factionalism and we all have like this distinct identity is this going to breed more war or is every state going to become so small because they'll just faction and keep dividing i mean you can get something the most homogenous human culture you can imagine and they will find something to divide them so so is this good is it bad if it we keep dividing is it going to be like a homogeneous like bunch of sub-factions or is it just going to lead to like more division and more war if i discuss that it will take a lot of time i understood this session was excellent we are much off track

definitely we'll save it for the dot one okay so moving on uh the collective mind does not directly emerge from inter-individual interactions but from slower evolving dynamical constraints which supervene on but do not reduce to human activity therefore the proper scale of analysis of collective cognition is an integrated activity of entire urban systems and should more specifically be grounded in the broader processes by which they maintain and reproduce their structural identity understood as an operationally closed set of constraints underlying their metabolic cognitive and ecological you know replace urban with brain body just you could say the same exact thing the brain the body have to be an integrated unit finally blue what does it mean to think like a state i'm going to put your overlay over the video so it'll look like you're a state go for it the cognitive activity of states is defined mechanically by the standardized administrative systems they use to understand and manipulate their world and functionally by their drive to build and enact grip over the material and sociocultural niche so there it is the question the answer we made it through from a to z and i like it's not what is it like to be a state it's not the nagel 54 what is it like to be a bat this is more of a cognitivist interpretation helps us actually potentially find useful ways you know i don't want to do a you know in in x you grip the state people can kind of write their own jokes on that theme nice um and we can just think about what maybe these questions lead to maybe what a good understanding of this work might enable um avail mentioned earlier the necessity for this understanding because of the the kind of divide divided space that we're in right now what are the next steps for the fep and what are the goals what are you still curious about if you want to participate in either of the discussion videos get in touch with us we would love to have you on to talk about these ideas in more detail do you guys have any closing thoughts yeah just in dot one let's dive in and open some threads take some notes in dot two let's see where we go maybe somebody would like to pick up on one of those threads or apply one of these ideas and say oh i was working on a history of my region and so i wanna see how i can apply that so we applied active inference to get thinking like a state and then how do we apply thinking like a state something else whatever it is that somebody is interested in seeing avel thank you for coming and kind of guiding us through our contextualization video it's always a pleasure do you want to have the final word i don't know and it is anything smart to say so yes but these are the words the paper does speak for itself uh quite brilliantly i will say all right authors who come on dot zero are the best no it's all good thanks for joining blue and facilitating see you later survival and thanks also to stephen so hope to see you all in the dot one with the dot two or beyond so bye bye